



US 20250270839A1

(19) **United States**

(12) **Patent Application Publication**  
**LIU**

(10) **Pub. No.: US 2025/0270839 A1**

(43) **Pub. Date: Aug. 28, 2025**

(54) **COLLAPSIBLE ARCED FLAGPOLE AND  
KIT FOR THE SAME**

**Publication Classification**

(51) **Int. Cl.**

**E04H 12/32** (2006.01)

**E04H 12/22** (2006.01)

(52) **U.S. Cl.**

**CPC** ..... **E04H 12/32** (2013.01); **E04H 12/2215**  
(2013.01)

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(21) Appl. No.: **19/059,636**

(22) Filed: **Feb. 21, 2025**

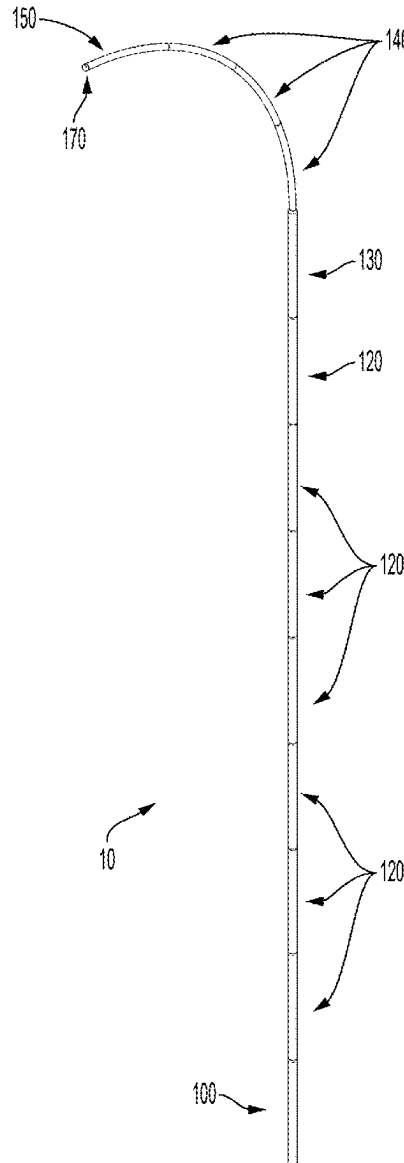
**Related U.S. Application Data**

(60) Provisional application No. 63/558,223, filed on Feb.  
27, 2024.

(57)

**ABSTRACT**

This specification describes a curved or arced collapsible flagpole for a feather flag. The collapsible flagpole is assembled from a kit comprising a plurality of straight hollow shafts, hollow arced shafts and a base shaft and tip shaft. Having the flagpole curved and not under tension from the flag is important as it makes the flag last longer.



Home → Custom Feather Flags → Hardware → Pole Kits

### Feather Flag Pole Kits

Each of our feather flag poles is made from a combination of aluminum and fiberglass poles that will bend to fit any of the shapes offered in their size range. You can use the same feather pole for the teardrop shape as you use for the straight, angled, concave and convex shapes. The feather flag poles are excellent for a variety of uses (both indoor and outdoor) and have been tested for a variety of wind speeds (up to 63 mph depending on the feather flag pole kit).

**Up to 63mph Wind-Rating | Huge Inventory | Same-day Shipping for Orders by 1pm ET**

Sort by: Position: Low to High ~

16 ~

|                                                                                                                                                                                                                |                                                                                                                                                                                       |                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Feather, Teardrop &amp; Rectangular<br/>(No Arm) Flag Pole Set<br/>☆☆☆☆☆<br/>17 Reviews<br/><b>\$ 18<sup>99</sup></b></p> |  <p>Rectangular Flag with Arm Pole<br/>Set<br/>☆☆☆☆☆<br/>0 Reviews<br/><b>\$ 51<sup>99</sup></b></p> |  <p>Razor Flag Pole Set<br/>☆☆☆☆☆<br/>0 Reviews<br/><b>\$ 76<sup>99</sup></b></p>     |
|  <p>Dart Flag Pole Set<br/>☆☆☆☆☆<br/>0 Reviews<br/><b>\$ 134<sup>99</sup></b></p>                                           |  <p>Crystal Flag Pole Set<br/>☆☆☆☆☆<br/>0 Reviews<br/><b>\$ 161<sup>99</sup></b></p>               |  <p>Surfer Flag Pole Set<br/>☆☆☆☆☆<br/>0 Reviews<br/><b>\$ 110<sup>99</sup></b></p> |

### Standard VS Premium Feather Flag Pole Kits

Display your feather flags outdoors in the grass or on the sidewalk with one of these feather flag pole kits. We include both standard and premium pole set options all of our feather flag shapes so that you can find the perfect one to suit your purpose. In this category, we include pole sets for all of our feather flag shapes: the standard feather flag kit, teardrop flags, rectangle flags with and without the arm, razor, dart, crystal, surfer, and moon.

**FIG. 1**  
PRIOR ART

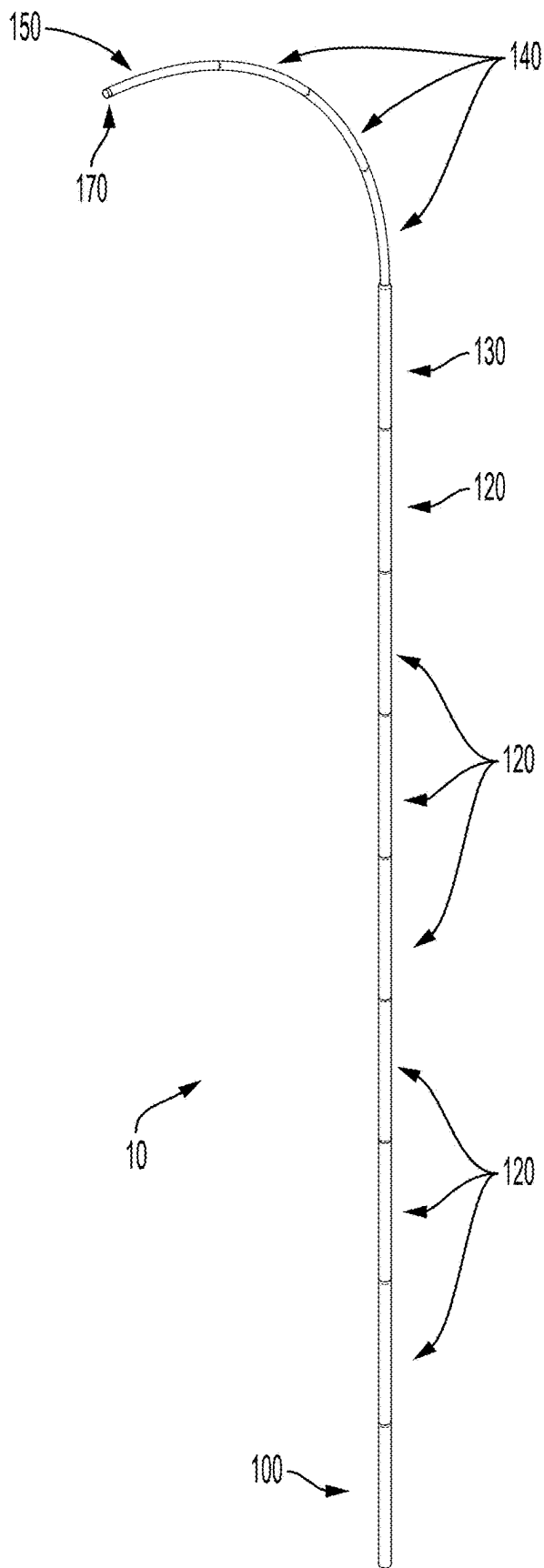
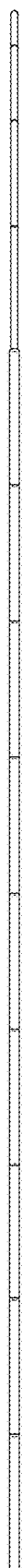


FIG. 2



10

FIG. 3



10

FIG. 4

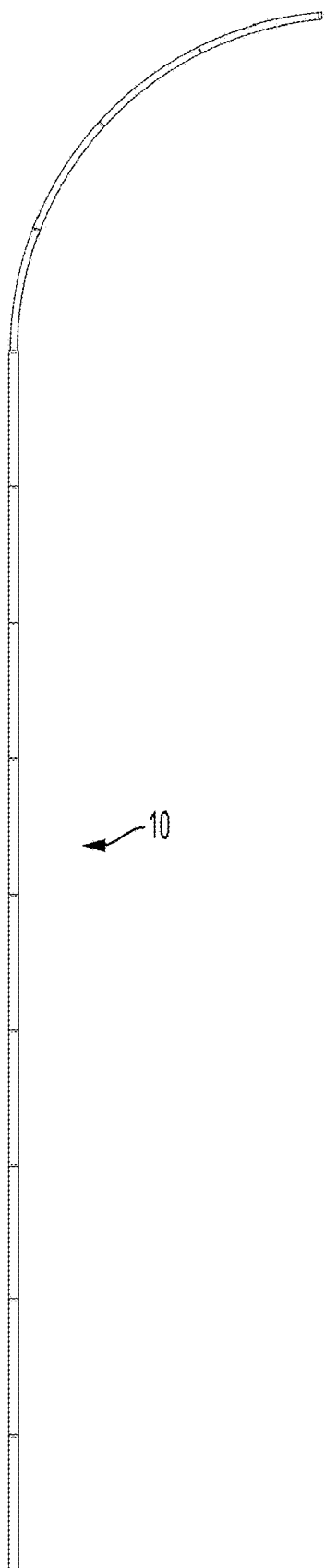


FIG. 5

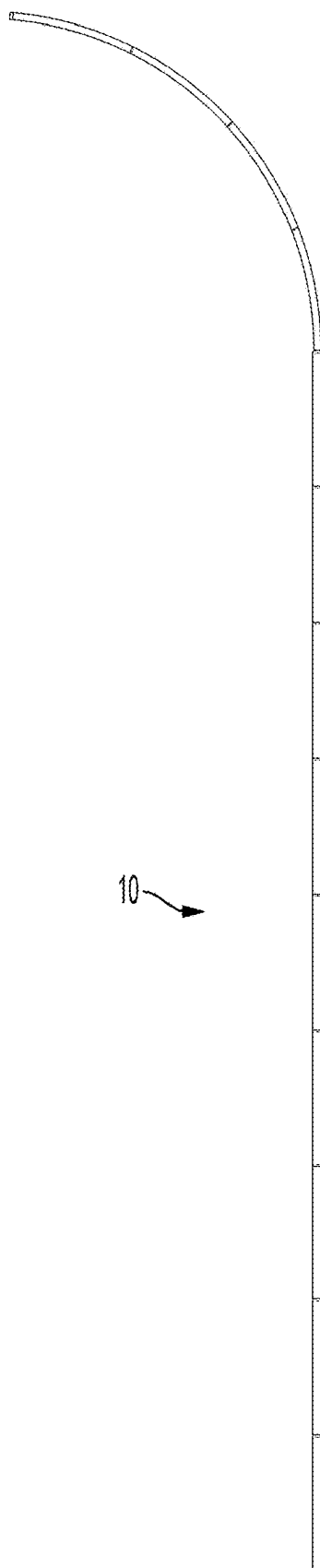


FIG. 6

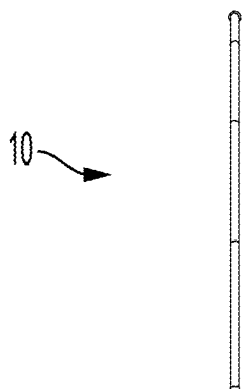


FIG. 7

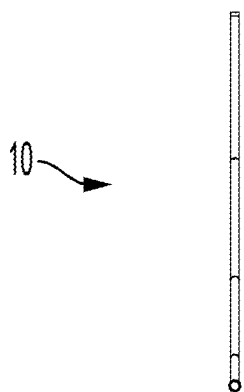


FIG. 8

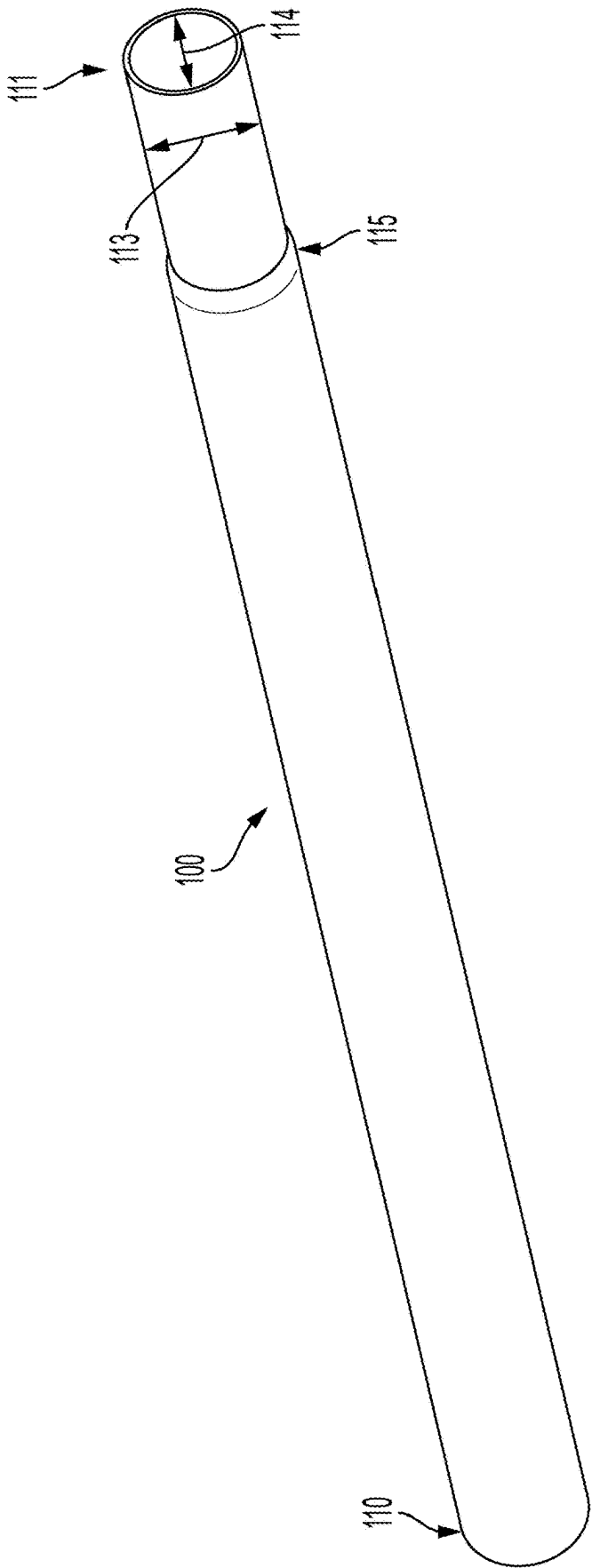


FIG. 9A

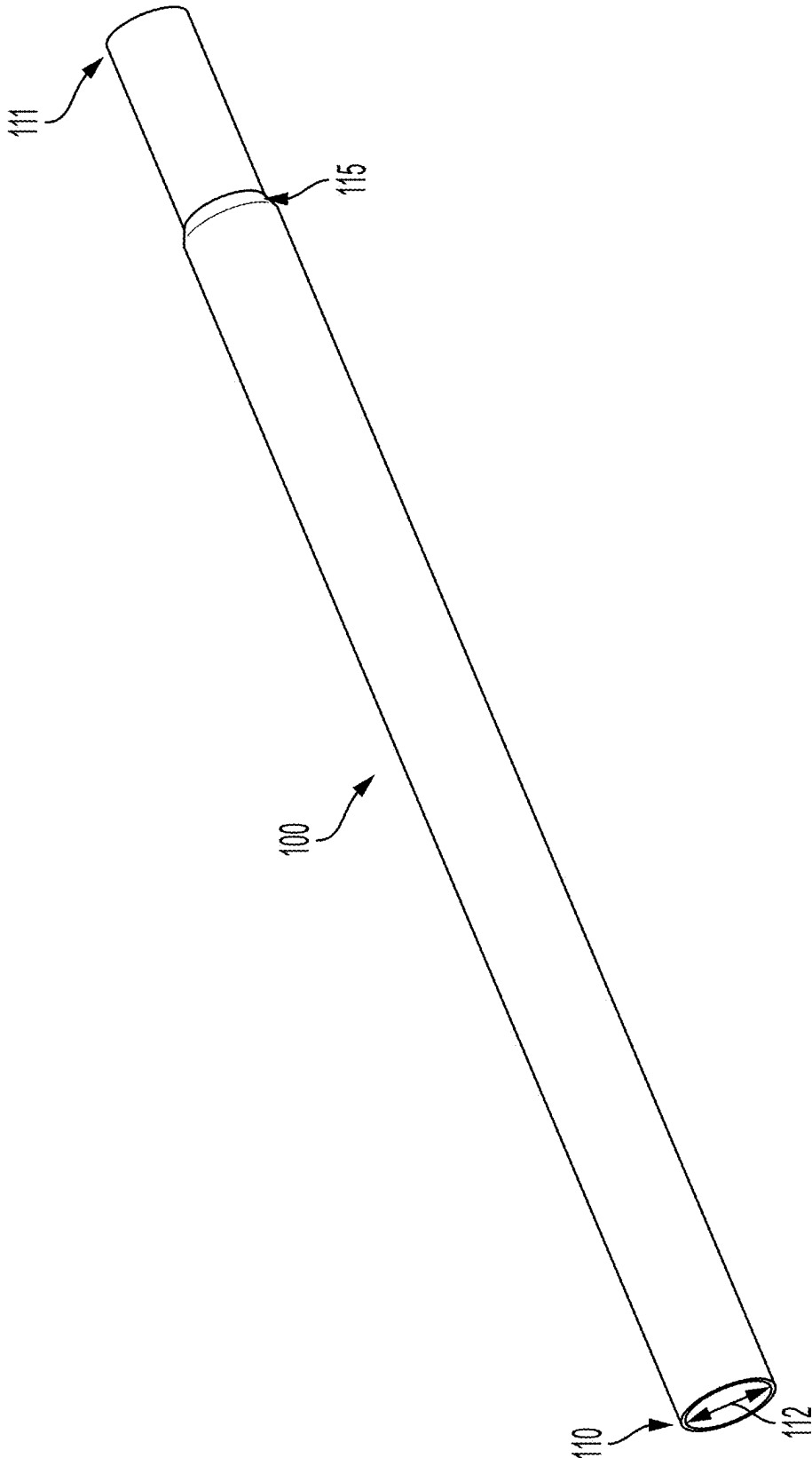


FIG. 9B



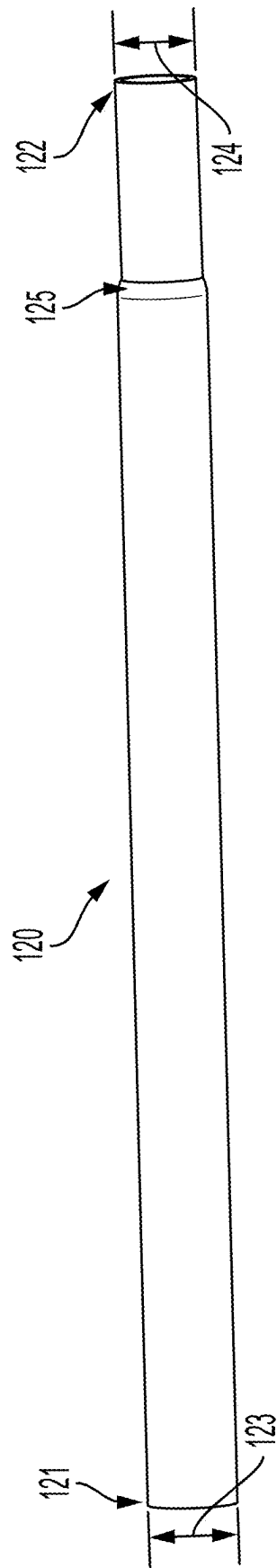


FIG. 10

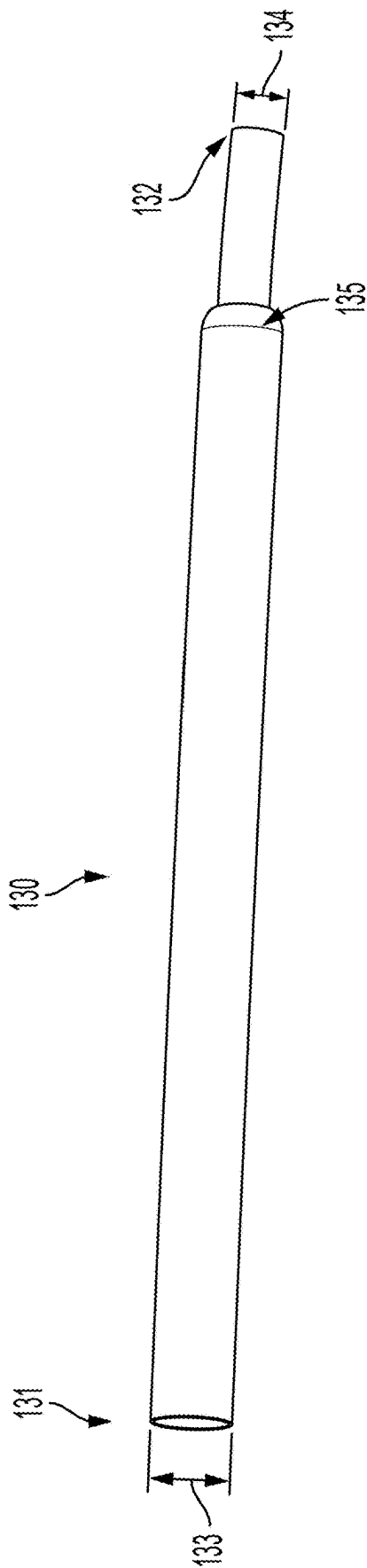


FIG. 11

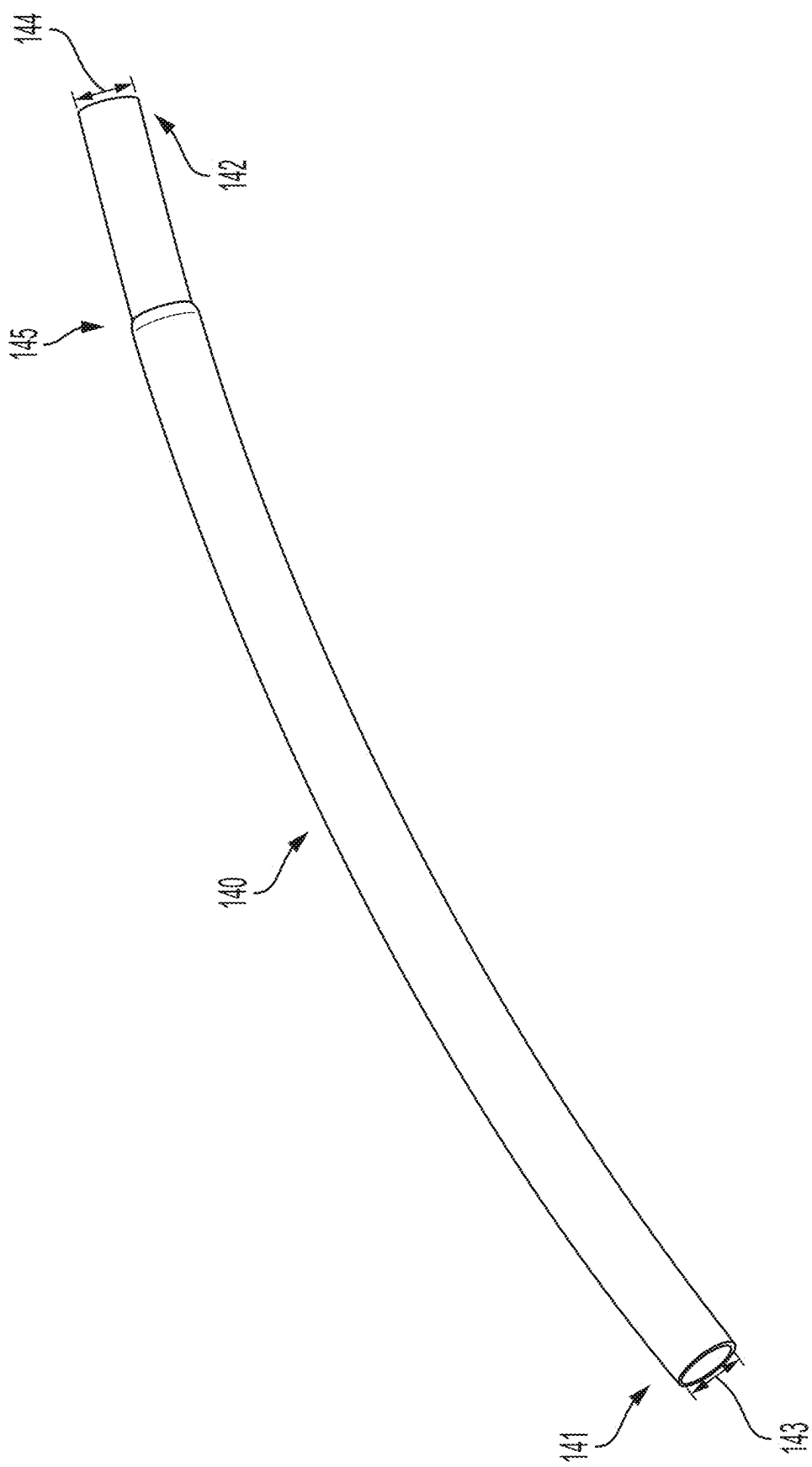


FIG. 12

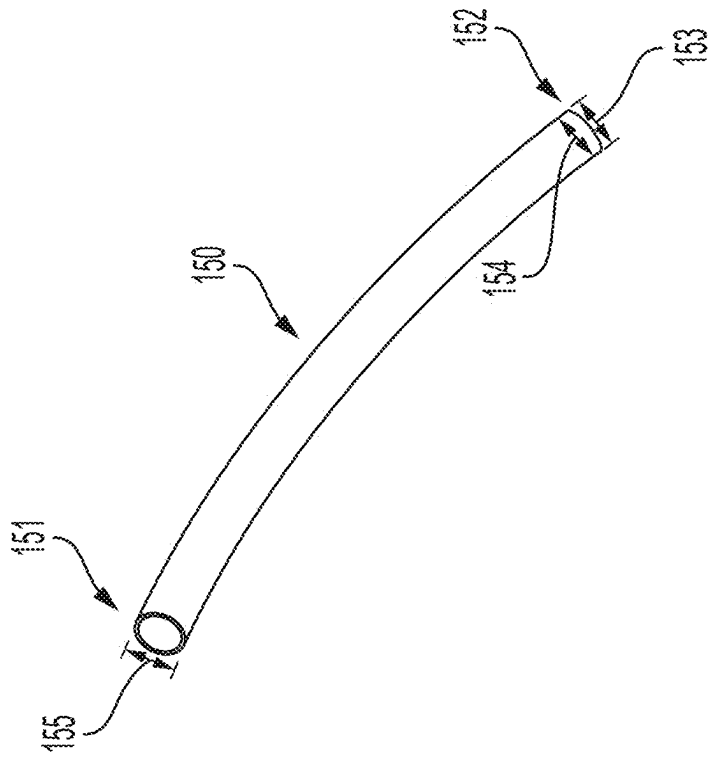


FIG. 13

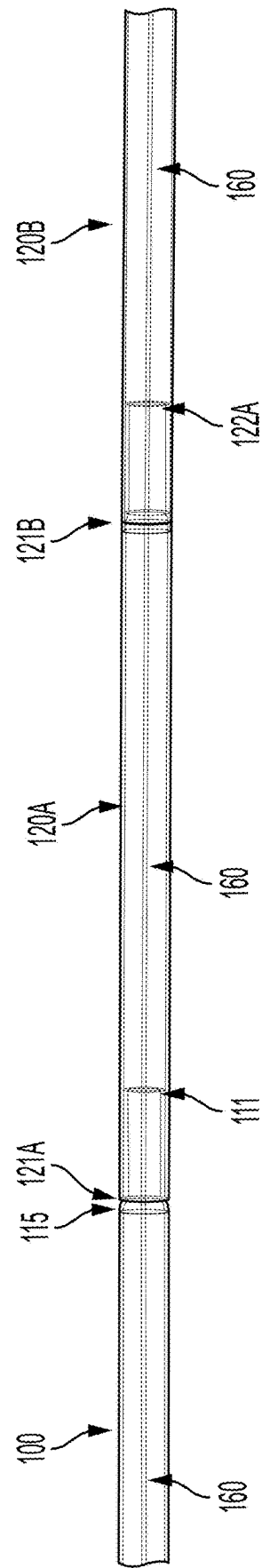


FIG. 14

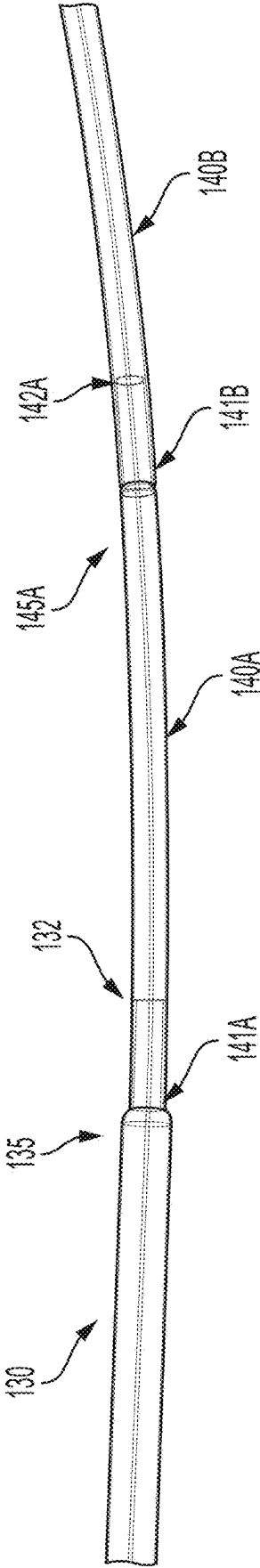


FIG. 15

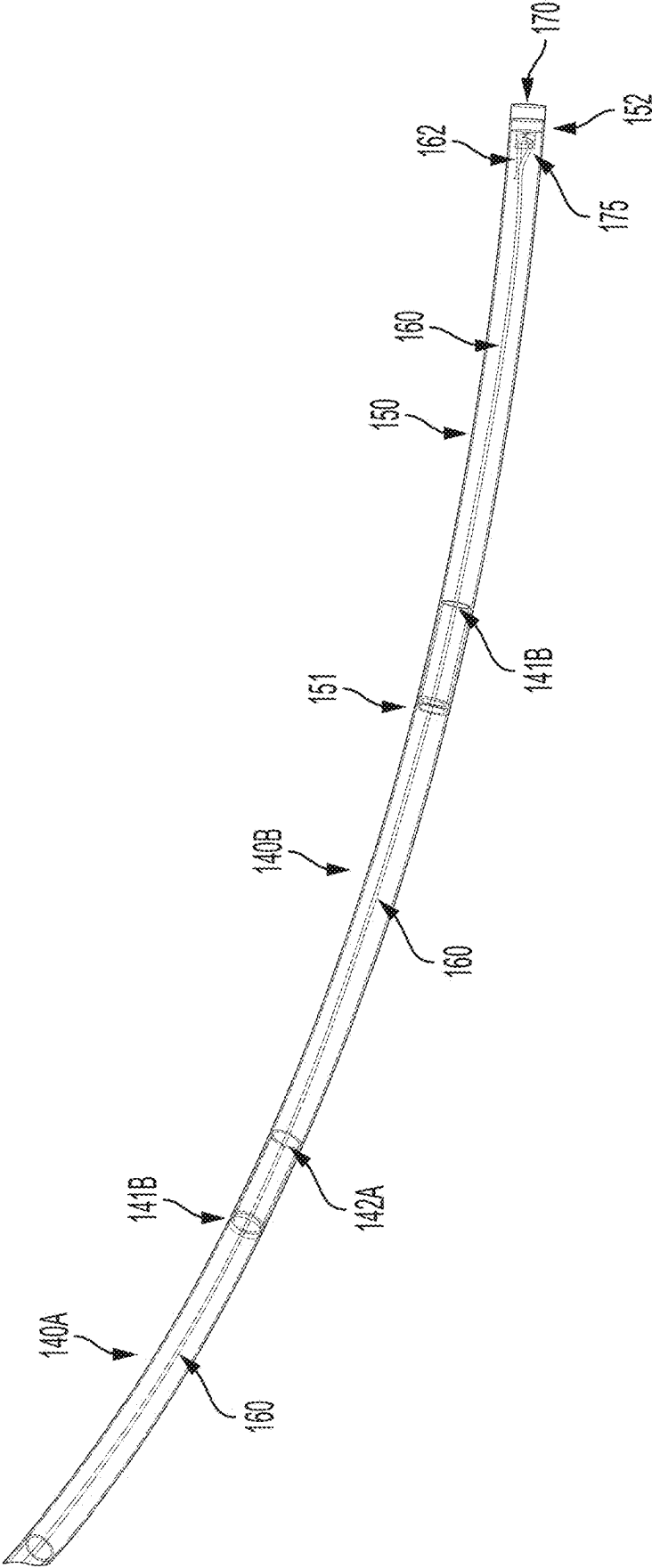


FIG. 16

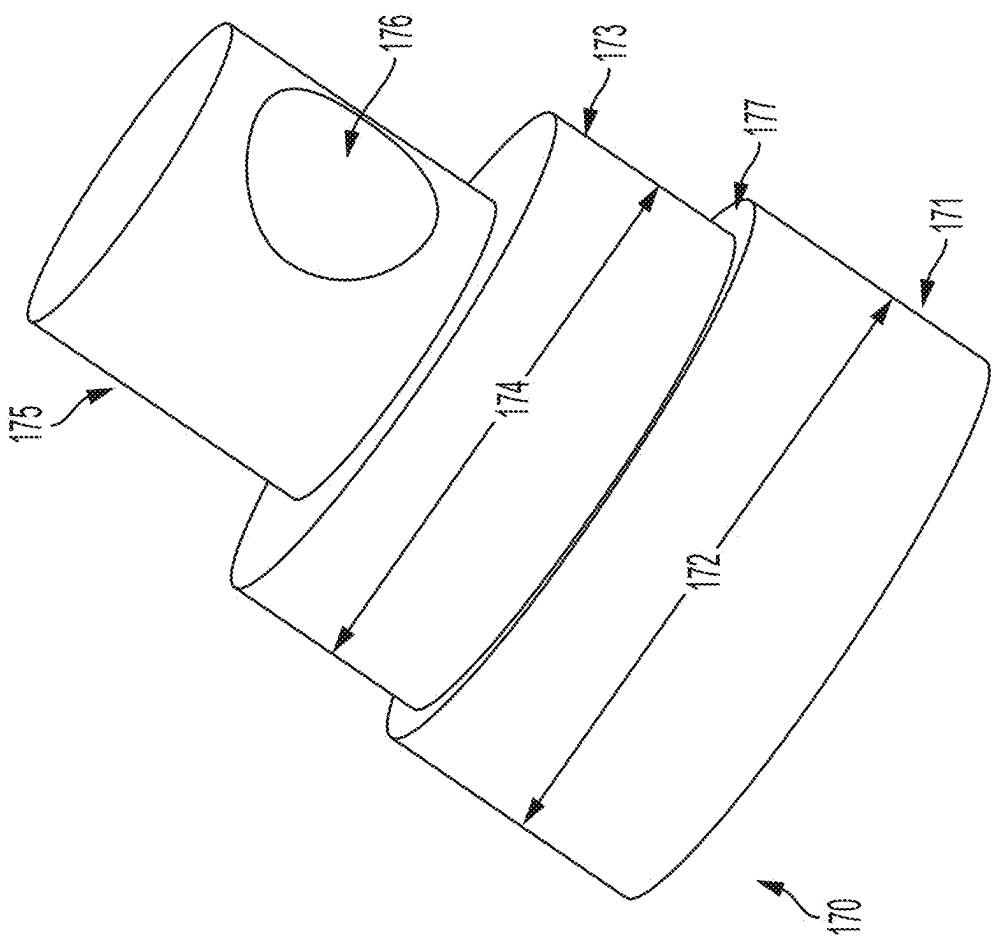


FIG. 17



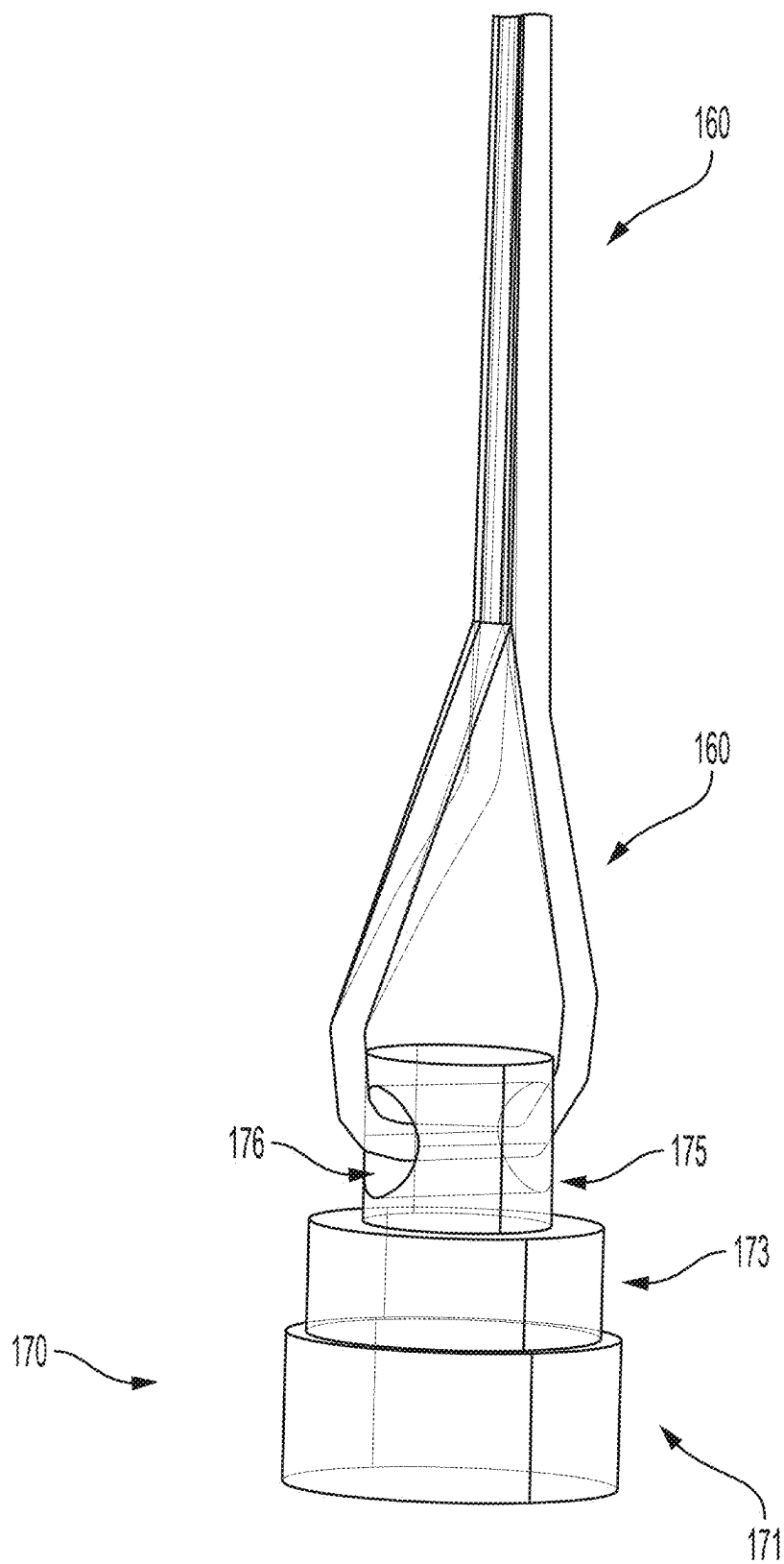


FIG. 18

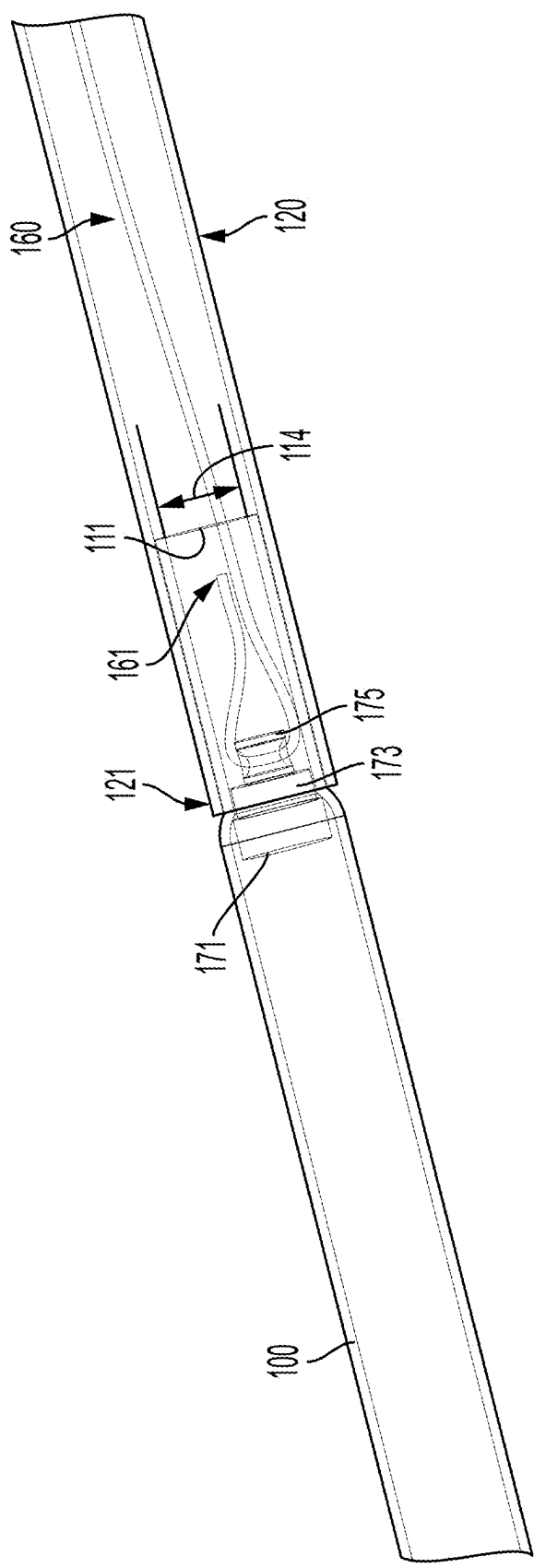


FIG. 19

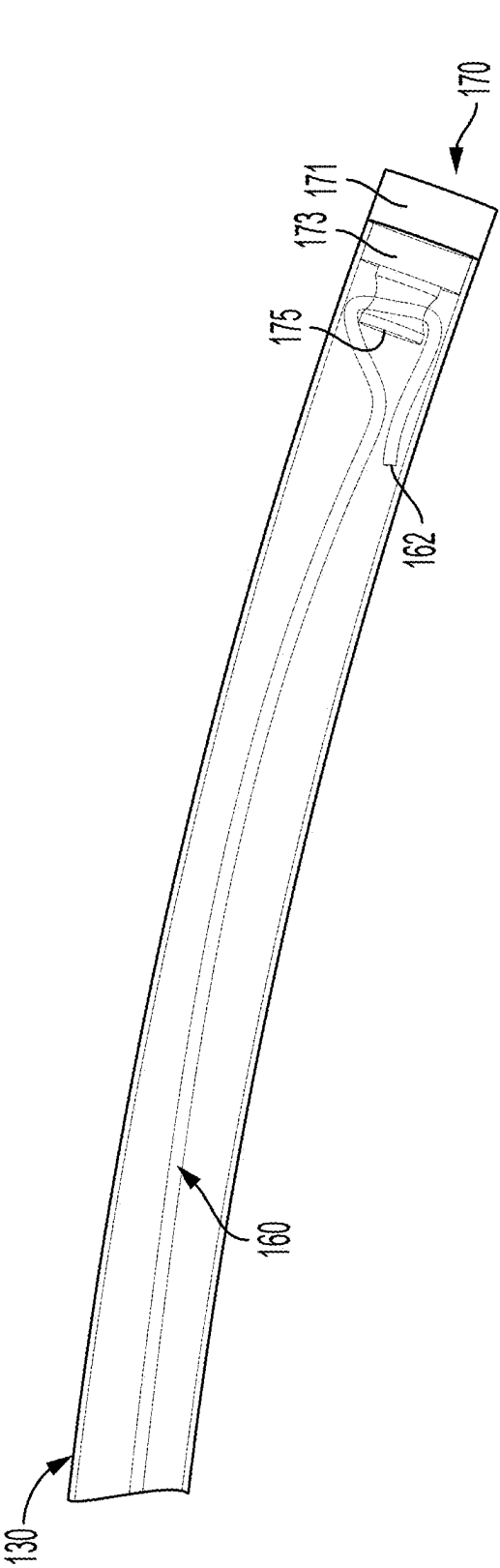


FIG. 20A

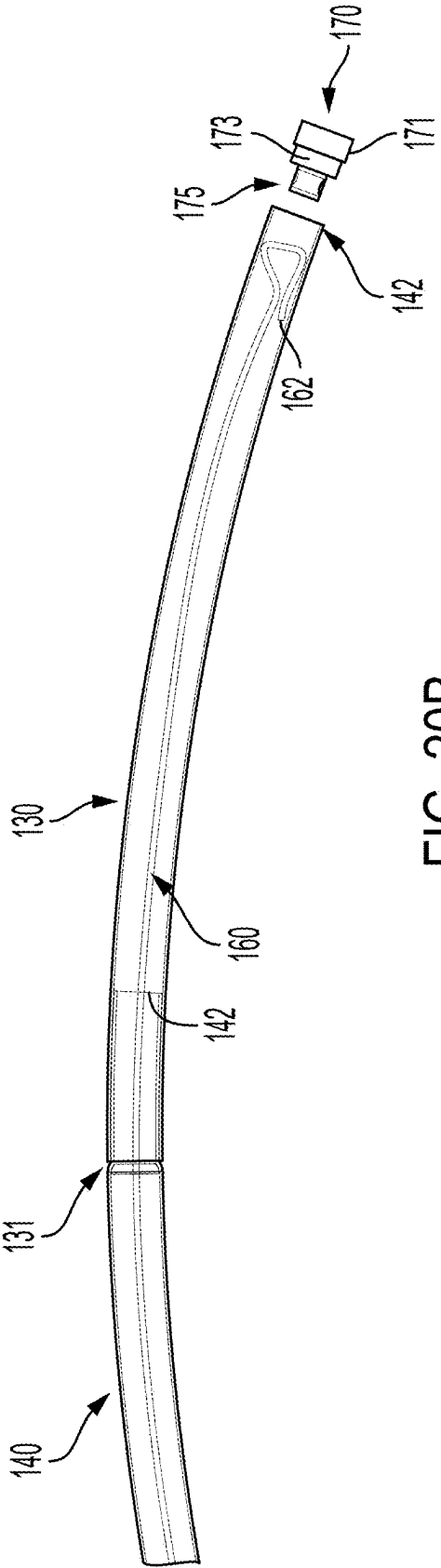
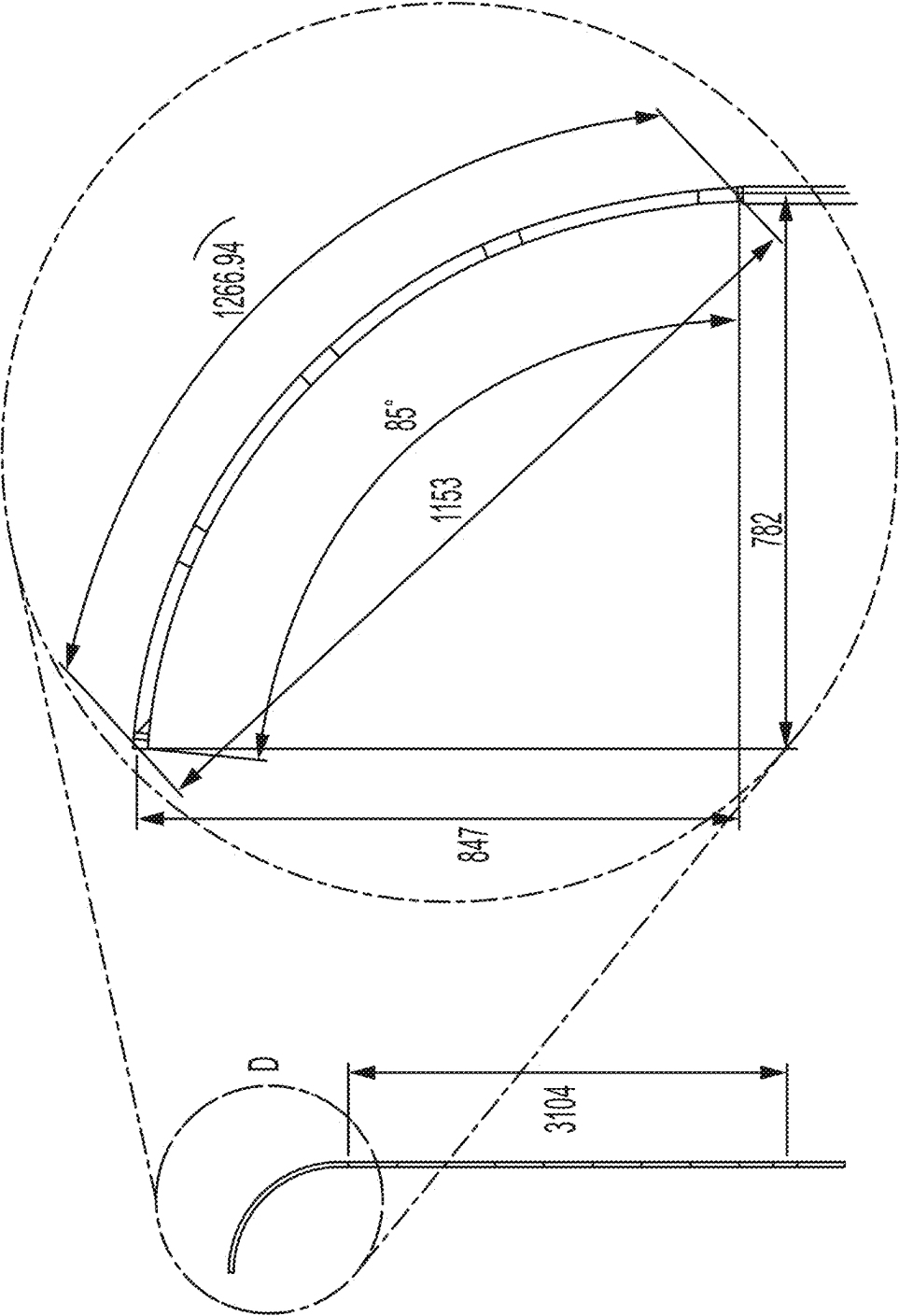


FIG. 20B



DETAIL D  
SCALE 1:10

FIG. 21

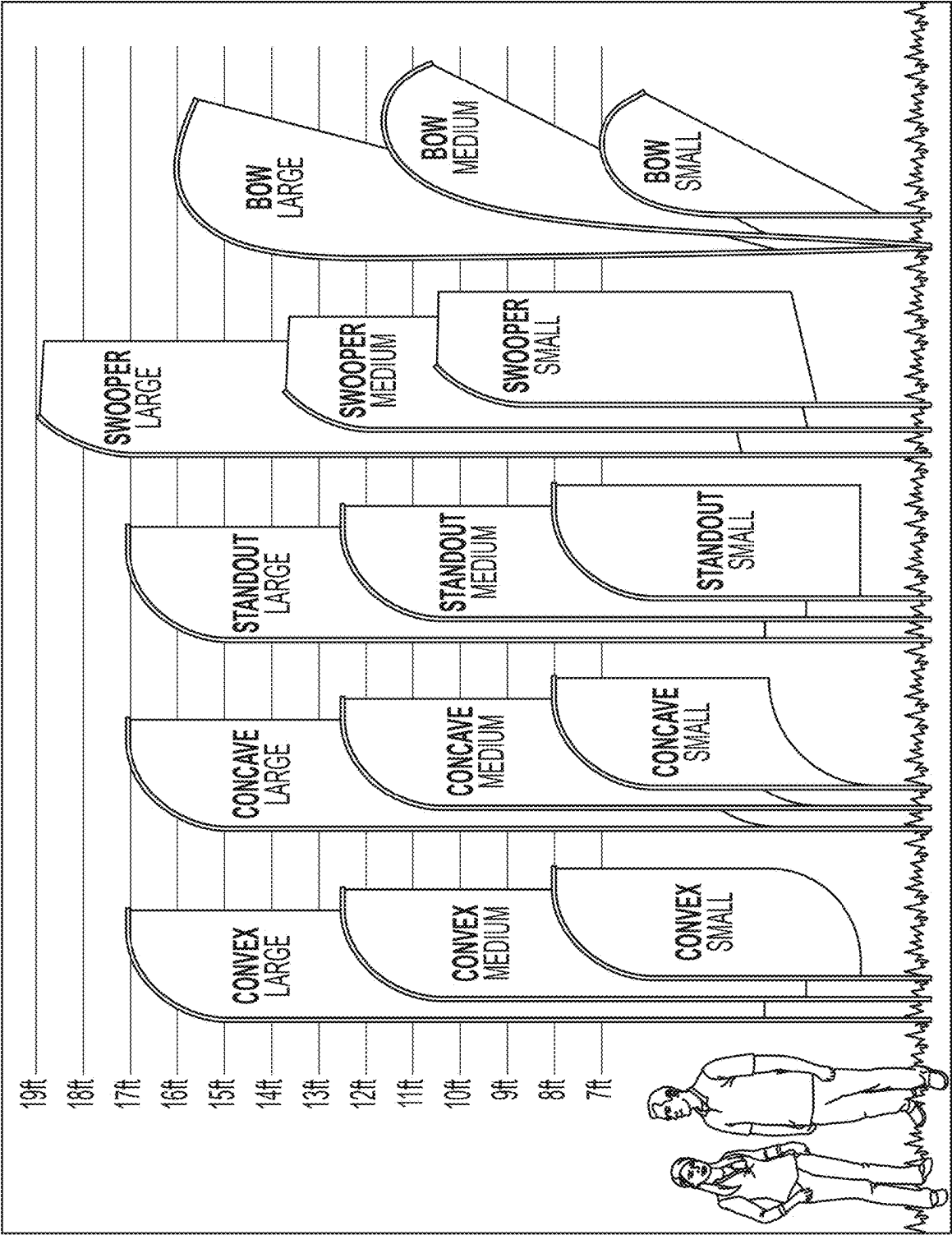


FIG. 22

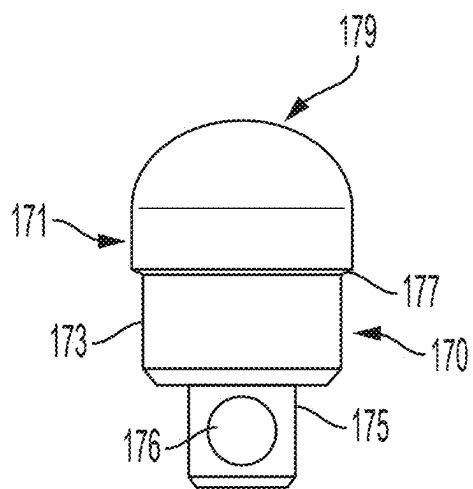


FIG. 23

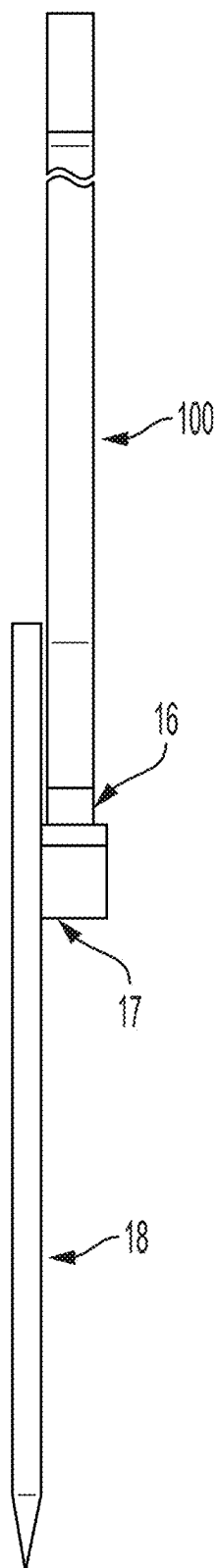


FIG. 24

## COLLAPSIBLE ARCED FLAGPOLE AND KIT FOR THE SAME

### CROSS REFERENCES AND PRIORITIES

**[0001]** This application claims priority from U.S. provisional patent application No. 63/558,223, filed 27 Feb. 2024, the teachings of which are incorporated herein in their entirety.

### BACKGROUND

**[0002]** Collapsible tent poles are known in the art. United States Patent Application 2009/0020146 to BONG, Kim published 22 Jan. 2009, (“BONG”) describes a series of hollow shafts, connected by insertion into each other, and an elastic cord arranged in the shafts with both ends [of the cord] secured to the two end most shafts, wherein at least one groove is arranged on at least one end of the shaft.

**[0003]** FIG. 3 of BONG shows that the un-assembled shafts are straight. Assembled FIG. 1 of BONG depicts the assembled shafts in a curve or arc that is formed by bending the otherwise straight shaft assembly and holding it in place by the tension of the tent.

**[0004]** Feather flags or Sail flags operate in a comparable manner.

**[0005]** As shown in FIG. 1 of this instant application the current commercial Feather Flagpole Kits are made from a combination of aluminum and fiberglass poles that will bend to fit any of the shapes offered in their size range. The web page of FIG. 1 further notes that one can use the same feather pole for the teardrop shape as one uses for the straight, angled, concave, and convex shapes.

**[0006]** The current prior art Feather Flagpole has a base shaft and each progressive shaft has a progressively smaller diameter. In this manner the thicker base shaft forms support while the thinner tip pole is allowed to bend as it follows the curved edge of the flag or banner fabric.

**[0007]** The problem encountered by the current prior art straight shaft flagpoles is that the assembled pole is bent by the flag creating a constant tension on the flag which causes the flag to eventually rip and tear prematurely.

### SUMMARY

**[0008]** This specification discloses a flagpole assembly comprised of assembly elements of a plurality of shafts with the plurality of shafts comprising a base shaft having a base shaft terminal end and a base shaft insert end opposite the base shaft terminal end.

**[0009]** It is further disclosed that the plurality of shafts have a connecting shaft having a connecting shaft insertion end and a connecting shaft insert end opposite the connecting shaft insertion end.

**[0010]** It is further disclosed that the plurality of shafts have tip shaft having a tip shaft insertion end and a tip shaft terminal end opposite the tip shaft insertion end.

**[0011]** It is further disclosed that the plurality of shafts have a plurality of straight shafts each having a straight shaft insertion end and a straight shaft insert end; and at least one arc shaft having an arc shaft insertion end and an arc shaft insert end.

**[0012]** It is further disclosed that the shafts are hollow and that the flagpole assembly further comprises a string passing through each shaft of the plurality of shafts with a string base

end attached to a first cap and a tip end of the string, opposite the base end of the string, attached to a second cap. This string may be elastic.

**[0013]** It is also disclosed that the base shaft insert end is configured to be inserted into at least one straight shaft insertion end, at least one straight shaft insert end is configured to be inserted into the connecting shaft insertion end, the connecting shaft insert end is configured to be inserted into an arc shaft insertion end, and at least one arc shaft with an arc shaft insert end configured to be inserted into a tip shaft insertion end.

**[0014]** It is further disclosed that the base shaft insert end may be inserted into at least one straight shaft insertion end, at least one straight shaft insert end is inserted into the connecting shaft insertion end and at least one arc shaft insert end is inserted into the tip shaft insertion end.

**[0015]** It is further disclosed that the first cap may comprise a first plug with a first plug diameter greater than the base shaft insert end inside diameter, a first cap insert end with a first cap insert diameter less than the base shaft insert end inside diameter, and a first cap string attachment.

**[0016]** It is also disclosed that the second cap may comprise a second cap plug with a second cap plug diameter greater than the tip shaft terminal end inside diameter, a second cap insert end with a second cap insert end diameter less than the tip shaft terminal end inside diameter, and a second cap string attachment member.

**[0017]** The specification also discloses that the string base end is attached to the first cap string attachment member.

**[0018]** The specification also discloses that the string tip end is attached to the second cap string attachment member.

**[0019]** It is further disclosed that the length of the flagpole assembly may be in the range of 7 feet to 19 feet.

**[0020]** The specification also discloses that the unassembled assembly elements may be grouped as a the kit which can fit into a box having internal dimensions of 18 inches long, 14 inches wide and 8 inches deep.

### BRIEF DESCRIPTION OF FIGURES

**[0021]** FIG. 1 is a prior art flagpole kit ready for assembly.

**[0022]** FIG. 2 is a perspective right hand view of an assembled embodiment of the flagpole.

**[0023]** FIG. 3 is a front view of an assembled embodiment of the flagpole.

**[0024]** FIG. 4 is a back view of an assembled embodiment of the flagpole.

**[0025]** FIG. 5 is a left view of an assembled embodiment of the flagpole.

**[0026]** FIG. 6 is a right view of an assembled embodiment of the flagpole.

**[0027]** FIG. 7 is a top view of an assembled embodiment of the flagpole.

**[0028]** FIG. 8 is a bottom view of an assembled embodiment of the flagpole.

**[0029]** FIG. 9A is a view of an exemplary base shaft rotated out of the page.

**[0030]** FIG. 9B is a view of an exemplary base shaft rotated into the page.

**[0031]** FIG. 10 is a view of an exemplary straight shaft.

**[0032]** FIG. 11 is a side view of an exemplary connecting shaft.

**[0033]** FIG. 12 is a perspective view of an exemplary arc shaft.



[0034] FIG. 13 is a perspective view of an exemplary tip shaft.  
 [0035] FIG. 14 is a view of a straight shaft assembled to another straight shaft.  
 [0036] FIG. 15 is a view of a connecting shaft assembled with an arc shaft.  
 [0037] FIG. 16 is a view of an arc shaft assembled with a tip shaft.  
 [0038] FIG. 17 is a view of a cap.  
 [0039] FIG. 18 is a view of a cap connected to a string.  
 [0040] FIG. 19 is a view of the cap with string attached and assembled with the base.  
 [0041] FIG. 20A is cross section view of the tip shaft with the cap and string installed.  
 [0042] FIG. 20B is an exploded cross section view of the tip shaft with the cap and string.  
 [0043] FIG. 21 demonstrates the calculation of the arc angle.  
 [0044] FIG. 22 depicts the various types of flags and flagpole assembly lengths or heights for the flagpole assembly.  
 [0045] FIG. 23 depicts an alternate embodiment of the cap and the end of the tip shaft.  
 [0046] FIG. 24 depicts the ground stake with the base mounted on it.

#### DETAILED DESCRIPTION

[0047] Disclosed herein is a flagpole for a feather flag. The flagpole is described below with reference to the Figures. As described herein and in the claims, the following numbers refer to the following assembly elements as noted in the Figures.

[0048] 10 refers to an assembled flagpole.  
 [0049] 16 refers to a socket.  
 [0050] 17 refers to a receiver.  
 [0051] 18 refers to a ground spike.  
 [0052] 100 refers to a base shaft.  
 [0053] 110 refers to a base shaft terminal end.  
 [0054] 111 refers to a base shaft insert end.  
 [0055] 112 refers to a base shaft terminal end inside diameter.  
 [0056] 113 refers to a base shaft insert outside diameter.  
 [0057] 114 refers to a base shaft insert inside diameter.  
 [0058] 115 refers to a base shaft shoulder.  
 [0059] 120 refers to a straight shaft.  
 [0060] 120A refers to the first straight shaft (A) connected to the base shaft.  
 [0061] 120B refers to the second straight shaft (B) connected to straight shaft (120A).  
 [0062] 121 refers to a straight shaft insertion end.  
 [0063] 121A refers to the straight shaft insertion end of the first straight shaft (A) connected to the base shaft.  
 [0064] 121B refers to the straight shaft insertion end of the second straight shaft (B) connected to the base shaft.  
 [0065] 122 refers to a straight shaft insert end.  
 [0066] 122A refers to the straight shaft insert end of the first straight shaft (A) connected to the base shaft.  
 [0067] 121A refers to the straight shaft insertion end of the first straight shaft (A) connected to the base shaft.  
 [0068] 122A refers to the straight shaft insert end of the first straight shaft (A) connected to the base shaft.  
 [0069] 123 refers to a straight shaft insertion end inside diameter.

[0070] 124 refers to a straight shaft insert end outside diameter.  
 [0071] 125 refers to a straight shaft shoulder.  
 [0072] 130 refers to a connecting shaft.  
 [0073] 131 refers to a connecting shaft insertion end.  
 [0074] 132 refers to a connecting shaft insert end.  
 [0075] 133 refers to a connecting shaft insertion end outside diameter.  
 [0076] 134 refers to a connecting shaft insert end outside diameter.  
 [0077] 135 refers to a connecting shaft shoulder.  
 [0078] 140 refers to an arc shaft.  
 [0079] 140A refers to a first arc shaft of an assembly.  
 [0080] 140B refers to a second arc shaft assembly assembled to first arc shaft 140A.  
 [0081] 141 refers to an arc shaft insertion end.  
 [0082] 141A refers to a first arc shaft insertion end for an assembly.  
 [0083] 141B refers to a second arc shaft insertion end for an arc shaft 140B assembled with first arc shaft 140A.  
 [0084] 142 refers to an arc shaft insert end.  
 [0085] 142A refers to an arc shaft insert end of first arc shaft 140A.  
 [0086] 143 refers to an arc shaft insertion end inside diameter.  
 [0087] 144 refers to an arc shaft insert end outside diameter.  
 [0088] 145 refers to an arc shaft shoulder.  
 [0089] 145A refers to an arc shaft shoulder of first arc shaft 140A.  
 [0090] 150 refers to a tip shaft.  
 [0091] 151 refers to a tip shaft insertion end.  
 [0092] 152 refers to a tip shaft terminal end.  
 [0093] 153 refers to a tip shaft terminal end outside diameter.  
 [0094] 154 refers to a tip shaft terminal end inside diameter.  
 [0095] 155 refers to a tip shaft insertion end inside diameter.  
 [0096] 160 refers to a string.  
 [0097] 161 refers to a string base end.  
 [0098] 162 refers to a string tip end.  
 [0099] 170 refers to a cap.  
 [0100] 171 refers to a cap plug.  
 [0101] 172 refers to a cap plug diameter.  
 [0102] 173 refers to a cap insert end.  
 [0103] 174 refers to a cap insert end diameter.  
 [0104] 175 refers to a cap string attachment member.  
 [0105] 176 refers to a cap string attachment member hole.  
 [0106] 177 refers to a cap plug rim.  
 [0107] 179 refers to a dome or hemispherical on top of the cap.

[0108] As shown in FIG. 2, the assembled flagpole assembly (10) is constructed from assembly elements which include a plurality of shafts. These shafts are typically hollow. The shafts are typically aluminum or fiberglass, but there could be both fiberglass shafts and aluminum shafts in the assembly.

[0109] As shown in FIG. 2, the first shaft, or bottom shaft, is called a base shaft (100).

[0110] The base shaft is connected to a series of straight shafts (120). There is a connecting shaft (130) that connects

to an arc shaft (140). The last shaft is called the tip shaft (150). The tip shaft has a cap (170) inserted at the terminal end of the tip shaft (152).

[0111] FIGS. 3 to 8 show various views of the assembled flagpole.

[0112] FIG. 9A and 9B show the base shaft (100) as it is rotated in and out of the page. The base shaft is typically hollow with a base shaft terminal end (110) and a base shaft insert end (111) opposite the base shaft terminal end. The base shaft terminal end will have a base shaft terminal end inside diameter (112). The base shaft insert end will have a base shaft insert outside diameter (113), a base shaft insert inside diameter (114) and a base shaft shoulder (115).

[0113] As shown in FIG. 10, the straight shaft (120) will have a straight shaft insertion end (121), a straight shaft insert end (122), a straight shaft insertion end inside diameter (123), a straight shaft insert end outside diameter (124) and a straight shaft shoulder (125).

[0114] As shown in FIG. 11, the connecting shaft (130) has a connecting shaft insertion end (131), a connecting shaft insert end (132) opposite the connecting shaft insertion end, a connecting shaft insertion end outside diameter (133), a connecting shaft insert end outside diameter (134), and a connecting shaft shoulder (135).

[0115] As shown in FIG. 12, the arc shaft (140) has an arc shaft insertion end (141), an arc shaft insert end (142), an arc shaft insertion end inside diameter (143), an arc shaft insert end outside diameter (144), and an arc shaft shoulder (145). An arc shaft is any shaft which is not straight. It could have one bend, multiple bends or a continuous bend from one end to the other.

[0116] As shown in FIG. 13 the tip shaft (150) has a tip shaft insertion end (151), a tip shaft terminal end (152) opposite the tip shaft insertion end, a tip shaft terminal end inside diameter (154), a tip shaft terminal end outside diameter (153) and a tip shaft insertion end inside diameter (155).

[0117] As shown in FIGS. 14-20B, there is a string (160) that passes through the hollow flagpole shafts. The string has a string base end (161) which is fixed inside the base shaft, and a string tip end (162) which is fixed inside the tip end.

[0118] As shown in FIGS. 17-20B, the string is attached to two caps, a base cap, which is the first cap which the first cap is inside the base shaft and a tip cap, which is the second cap which is inside the tip shaft. As shown in FIG. 17, the cap (170) has a cap plug (171), a cap plug diameter (172), a cap insert end (173), a cap insert end diameter (174), a cap string attachment member (175), a cap string attachment member hole (176), and a cap plug rim (177). The cap plug is shown as having a disk shape. An alternative embodiment is shown in FIG. 23 which displays a semi-hemispherical shape (179) at the end of the cap, making the exposed part of the cap look like a dome.

[0119] The flagpole is then assembled according to the above elements. The base shaft insert end is configured to be inserted into at least one straight shaft insertion end. An example of this configuration is shown in FIG. 14. Base shaft (100) is configured to be attached to the first straight shaft (120A) by making the base shaft insert end outside diameter slightly smaller than the straight shaft insertion end inside diameter. This allows the base shaft insert end to enter the straight shaft insertion end. It is preferable the diameters be selected so that the base shaft insert end can be frictionally engaged with the base shaft insert end. The selection

that the insertion end diameter is slightly larger than the corresponding insert end is consistent with all the shafts. The preferable frictional engagement is also preferable for all shaft-to-shaft connections.

[0120] As also shown in FIG. 14, straight shaft (120A) is connected with the second straight shaft (120B) with the insert end (122A) of straight shaft 120A being inserted into the straight shaft insertion end (121B) of straight shaft 120B. As stated earlier, this embodiment can be assembled because the insert end of one shaft is smaller than the inside diameter of the insertion end of the shaft into which the insert end is inserted.

[0121] The series of straight shafts will each have a straight shaft insertion end to receive the insert end of another shaft and a straight shaft insert end to be inserted into the end of yet another shaft.

[0122] At the end of the straight shafts is the connecting shaft (130). As shown in FIG. 15, the connecting shaft (130) connects to first arc shaft (140A) with the insertion of the connecting shaft insertion end into the first arc shaft insertion end (141A). As can be seen in FIG. 15, the insertion end outside diameter of the connecting shaft (FIG. 11, at 133) is significantly larger than the outside diameter of the arc shaft insertion end (141A). The ratio of the arc shaft insertion end outside diameter to the connecting shaft insertion end outside diameter is preferably less than 1, with 0.9 being a more preferred ratio, with 0.8 being even more preferred, 0.75 being even more preferred and a ratio of less than 0.73 being more preferred and less than 0.70 being even more preferred. It has been surprisingly discovered that the smaller diameter of the arc shaft outside diameter is highly functional, as the reduced diameter allows for compatibility with a wider range of flag sleeves, making the entire flagpole set more versatile and suitable for a 'universal fit.'

[0123] As further shown in FIG. 15, the first arc shaft (140A) is connected in a similar manner to the next arc shaft, a second arc shaft (140B). There could be one arc shaft or a plurality of arc shafts depending upon the amount of arc desired and over what amount of distance.

[0124] FIG. 16 shows the connection of one arc shaft (140A) with arc shaft (140B) to tip shaft (150). The tip shaft is the last shaft of the assembly. The tip shaft has a tip shaft insertion end (151) to receive the insert end of a preceding arc shaft (141B) and a tip shaft terminal end (152) to receive the cap (170). Also shown is string (160) passing through the hollow shafts and secured at the tip shaft terminal end (152) with the cap (170).

[0125] As seen in FIG. 21, the curved assembly has an arc. The arc length is 1266.94 mm. The straight part of the Assembly has a length of 3104 mm. In this Figure, the arc length is 1266.94 mm and the radius of the circle being 847 mm. The arc of 85 degrees in this case is determined by dividing the arc length (1267) by the radius (847 mm) and multiplying the result by 180 and dividing by  $\text{Pi}(\pi)$ .

[0126]  $1266.94/847*180/\pi=85.64$  Degrees in this case.

[0127] This arc is typically in the range of 30 to 90 Degrees.

[0128] While the tip shaft (150) is shown as curved having an arc, it is not essential. If the tip shaft has an arc then the arc angle defined by the flag could be made from one arc shaft and a tip shaft or a plurality of arc shafts and the tip shaft.

[0129] As noted throughout, the shafts are hollow with a string secured at the base shaft, and passing through the

straight shafts, the curved shafts, and secured at the tip shaft. One way to secure the string at the tip shaft is to attach the string to a tip plug. Although not necessary, the string is preferably elastic and provides tension through the hollow shaft that help keep the insert end interacted with an insertion end. The string preferably provides tension through the hollow shafts that help keep the insert end interacted with an insertion end. The string is preferably under tension before assembly and/or after assembly.

**[0130]** As shown in FIG. 19, the string is secured at the base shaft (100). In this embodiment it is secured by attaching the string with a cap. This is done by making the diameter of the cap plug (171) smaller than the inside diameter of the base shaft until it reaches the insert end which has an insert diameter (114) less than the diameter of the cap plug (171), but larger than the insert diameter. The diameter of the cap insert end (173) is smaller than the insert diameter of the base shaft which allows the string to move through the base shaft insert end. The string is affixed to the cap at the cap string attachment member (175) by passing the string through the hole (176) and affixing the string to itself, which can be done in many ways such as melting or gluing or tying. There are many ways to design the plug or cap and the cap in the base shaft does not need to be the same design as the cap at the tip end.

**[0131]** The string is secured at the tip shaft. FIGS. 20A and 20B show an example of how the string is secured at the tip shaft. The string is attached to the cap as described earlier. In this case, the diameter of the cap plug (171) is the same or greater than the outside diameter of the tip shaft terminal end (152). The diameter of the cap insert end (173) is slightly smaller than the inside diameter of the tip shaft terminal end which allows the cap insert end to be inserted into the terminal end of the tip shaft as shown in FIG. 20A. Again, the outer diameter of the cap insert end should be slightly smaller than the corresponding insertion end. Preferably, there is a friction force between the insert end and the insertion end that requires a force greater than the force of gravity to separate the shafts. A frictional relationship of the cap insert end with the inside of the tip shaft is preferred. However, because the string is preferably elastic, it provides the tension upon the shaft components to keep the flagpole assembled.

**[0132]** In windy conditions, the frictional connections may not be preferred as the elastic cord would allow the individual shaft members to move and adjust slightly without bending or breaking. Therefore, a flagpole that is void of frictional connection with a succeeding or preceding shaft member is an embodiment of this invention.

**[0133]** As shown in FIG. 22 the length, which is also the height, of the flagpole assembly can be in the range of 7 feet to 19 feet. With the range of 8 feet to 17 feet more preferred and 9 feet to 14 feet the most preferred.

**[0134]** In one embodiment the diameter of the tip shaft diameter be 18 mm or less. The diameter of the straight shafts by comparison are at least 30% greater than the diameter of the arced shaft.

**[0135]** In one embodiment, the base shaft and at least one straight shaft will have a wall thickness of at least 2 mm, with the arced shafts having a wall diameter of 1 mm to reduce material cost. By making the lower shafts thicker, the flagpole assembly is more resistant to wind gusts and does not bend as easily.

**[0136]** The base shaft should be strong enough withstand high winds. It has been observed that flagpole failures occur at the base shaft and can be solved by designing the base shaft so that it take at least 99N of force to make a fixed bend in the base shaft.

**[0137]** A standard flagpole mount can be used to mount the assembled flagpole into the ground. This standard mount is shown in FIG. 24 and has a ground spike (18) on one end which is inserted into the ground and a receiver (17) for receiving a socket (16) with the base shaft (100) fitting over the socket and thus fixing the flagpole perpendicular to the ground.

**[0138]** This flagpole assembly can be provided as a kit for assembling into the flagpole assembly. This kit comprises the assembly elements and all of the assembly elements of the kit can fit together into a box having internal dimensions of 18 inches long, 14 inches wide and 8 inches deep before assembling them. The assembly elements before being assembled are known as unassembled assembly elements. This is done for more economical shipping.

What I claim is:

1. A kit for assembling into a flagpole assembly, said kit comprising unassembled assembly elements of:

- a plurality of shafts;
- the plurality of shafts comprising
  - a base shaft (100) having a base shaft terminal end (110) and a base shaft insert end (111) opposite the base shaft terminal end;
  - a plurality of straight shafts (120) each having a straight shaft insertion end (121), and a straight shaft insert end (122) opposite the straight shaft insertion end;
  - a connecting shaft (130) having a connecting shaft insertion end (131), a connecting shaft insertion end outside diameter (133), and a connecting shaft insert end (132) opposite the connecting shaft insertion end;
  - at least one arc shaft (140) having an arc shaft insertion end (141), an arc shaft insertion end outside diameter and an arc shaft insert end (142) after the arc shaft insertion end; and
  - a tip shaft (150) having a tip shaft insertion end (151) and a tip shaft terminal end (152) opposite the tip shaft insertion end.

2. The kit of claim 1, wherein a ratio of the arc shaft insertion end outside diameter to the connecting shaft insertion end outside diameter is less than 1.

3. The kit of claim 1, further comprising a string (160) having a string base end secured at the base shaft and a string tip end secured at the tip shaft, wherein the string (160) passes through the plurality of straight shafts, the connecting shaft, and the at least one arc shaft.

4. The kit of claim 3, wherein the string is under tension after the flagpole assembly is assembled.

5. The kit of claim 3, wherein the string is secured at the base shaft with a first cap comprising a first plug (171) with a first plug diameter (172) greater than the base shaft insert end inside diameter (114), a first cap insert end (173) with a first cap insert end diameter (174) less than the base shaft insert end inside diameter, and a first cap string attachment member (176).

6. The kit of claim 4, wherein the string is secured at the base shaft with a first cap comprising a first plug (171) with a first plug diameter (172) greater than the base shaft insert end inside diameter (114), a first cap insert end (173) with

a first cap insert end diameter (174) less than the base shaft insert end inside diameter, and a first cap string attachment member (176).

7. The kit of claim 5, wherein the string is secured at the tip shaft with a second cap comprising a second cap plug with a second cap plug diameter greater than the tip shaft terminal end inside diameter, a second cap insert end with a second cap insert end diameter less than the tip shaft terminal end inside diameter, and a second cap string attachment member.

8. The kit of claim 6, wherein the string is secured at the tip shaft with a second cap comprising a second cap plug with a second cap plug diameter greater than the tip shaft terminal end inside diameter, a second cap insert end with a second cap insert end diameter less than the tip shaft terminal end inside diameter, and a second cap string attachment member.

9. The kit of claim 7, wherein the string base end is attached to the first cap string attachment member.

10. The kit of claim 8, wherein the string tip end is attached to the second cap string attachment member.

11. The kit of claim 1, wherein the unassembled assembly elements can fit together into a box having internal dimensions of 18 inches long, 14 inches wide and 8 inches deep.

12. The kit of claim 1 wherein the base shaft insert end is configured to be inserted into at least one straight shaft insertion end, at least one straight shaft insert end is configured to be inserted into the connecting shaft insertion end, the connecting shaft insert end is configured to be inserted into the arc shaft insertion end of the at least one arc shaft, and the at least one arc shaft with the arc shaft insert end configured to be inserted into the tip shaft insertion end.

13. The kit of claim 12, wherein a ratio of the arc shaft insertion end outside diameter to the connecting shaft insertion end outside diameter is less than 1.

14. The kit of claim 12, wherein the kit further comprises a string (160) secured at the base shaft and secured at the tip

shaft, wherein the string (160) passes through the plurality of straight shafts, connecting shaft, and the at least one arc shaft.

15. The kit of claim 14, wherein the string is under tension after the flagpole assembly is assembled.

16. The kit of claim 14, wherein the string is secured at the base shaft with a first cap comprising a first plug (171) with a first plug diameter (172) greater than the base shaft insert end inside diameter (114), a first cap insert end (173) with a first cap insert end diameter (174) less than the base shaft insert end inside diameter, a first cap string attachment member with the string is attached to the first cap string attachment member (176); and

wherein the string is secured at the tip shaft with a second cap comprising a second cap plug with a second cap plug diameter greater than the tip shaft terminal end inside diameter, a second cap insert end with a second cap insert end diameter less than the tip shaft terminal end inside diameter, a second cap string attachment member and the string is attached to the second cap string attachment member.

17. The kit of claim 12, wherein the unassembled assembly elements can fit together into a box having internal dimensions of 18 inches long, 14 inches wide and 8 inches deep.

18. The kit of claim 1, assembled into the flagpole assembly wherein the base shaft insert end is inserted into at least one straight shaft insertion end, at least one straight shaft insert end is inserted into the connecting shaft insertion end and at least one arc shaft insert end is inserted into the tip shaft insertion end.

19. The kit of claim 18 assembled into the flagpole assembly wherein a length of the flagpole assembly is in a range of 7 feet to 19 feet.

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